Compact, Organized Overview – Probability & Statistics Homework 3

Problem 1 – Biased Coin (Binomial)

- Model: $X \sim \text{Binomial}(n = 10, p = 0.3)$
- Key probabilities:
- $P(X = 0) = 0.7^{10} \approx 0.0283$
- $P(X = 3) = \binom{10}{3}(0.3)^3(0.7)^7 \approx 0.266$
- $P(X \geq 1) = 1 - 0.7^{10} \approx 0.9718$
- Moments: $E[X] = 3$, $\text{Var}(X) = 2.1$

Problem 2 – Defective Bulbs (Binomial)

- Model: $Y \sim \text{Binomial}(50, 0.02)$
- Key probabilities:
- $P(Y = 0) = 0.98^{50} \approx 0.364$
- $P(Y \leq 2) \approx 0.922$
- Moments: $E[Y] = 1$, $\text{Var}(Y) = 0.98$, $\text{SD}(Y) \approx 0.99$

Problem 3 – Binomial CDF Interpretation

- $Z \sim \text{Binomial}(8, 0.6)$
- CDF as sum: $P(Z \leq 3) = \sum_{k=0}^{3} \binom{8}{k}(0.6)^k(0.4)^{8-k}$
- Interpretation: "at most 3 successes in 8 trials".
- If $P(Z \leq k) = 0.9$, then k is the 90th percentile (90% quantile) of this binomial distribution.

Problem 4 – Poisson Calls ($\lambda = 4$)

- Model: $N \sim \text{Poisson}(4)$
- Key probabilities:
- $P(N = 0) = e^{-4} \approx 0.0183$
- $P(N = 1) = 4e^{-4} \approx 0.0733$
- $P(N \geq 2) \approx 0.9084$
- Moments: $E[N] = 4$, $\text{Var}(N) = 4$

Problem 5 – Poisson Approximation to Binomial (Clicks)

- Exact: $X \sim \mathrm{Binomial}(10000, 0.0005)$
- Approximate: $X \approx \mathrm{Poisson}(\lambda = 5)$
- Key approximations:
- $P(X = 0) \approx e^{-5} \approx 0.00674$
- $P(X \geq 1) \approx 0.99326$

Problem 6 – Sum of Poisson RVs

- $X_1 \sim \mathrm{Poisson}(2)$, $X_2 \sim \mathrm{Poisson}(3)$, independent
- Sum: $T = X_1 + X_2 \sim \mathrm{Poisson}(5)$
- Key probability:
- $P(T = 4) = e^{-5} \frac{5^4}{4!} \approx 0.175$
- Moments: $E[T] = 5$, $\mathrm{Var}(T) = 5$

Problem 7 – Standard Normal $Z \sim N(0, 1)$

- Use CDF $\Phi(z) = P(Z \leq z)$
- Forms:
- $P(Z \leq 1.2) = \Phi(1.2)$
- $P(Z > -0.5) = \Phi(0.5)$
- $P(-1 < Z < 2) = \Phi(2) - \Phi(-1) = \Phi(2) + \Phi(1) - 1$

Problem 8 – Normal with $\mu = 50, \sigma^2 = 9$

- Model: $X \sim N(50, 9)$, so $\sigma = 3$
- Standardization: $Z = (X - 50)/3$
- Key probabilities:
- $P(X \leq 47) = \Phi(-1) \approx 0.1587$
- $P(46 < X < 55) = \Phi(5/3) - \Phi(-4/3)$ (≈ 0.86)
- $P(X > 50) = 0.5$

Problem 9 – Heights $H \sim N(170, 8^2)$

- z-score for 186 cm: $z = 2$
- Key probabilities/quantiles:
- $P(H > 186) = 1 - \Phi(2) \approx 0.0228$
- 90th percentile: $h = 170 + 8z_{0.9} \approx 170 + 8(1.28) \approx 180.2\,\mathrm{cm}$

Problem 10 – Normal Approx to Binomial $X \sim \mathrm{Bin}(100, 0.4)$

- Moments: $E[X] = 40$, $\mathrm{Var}(X) = 24$, $\sigma \approx 4.90$
- Normal approx: $X \approx N(40, 24)$
- Continuity correction for $P(X \leq 35)$:
- Approximate $P(X \leq 35) \approx P(Y \leq 35.5)$ for $Y \sim N(40, 24)$
- $z \approx (35.5 - 40)/\sqrt{24} \approx -0.92$
- $P(X \leq 35) \approx \Phi(-0.92) \approx 0.18$

Problem 11 – Comparing Models

- $X_1 \sim \mathrm{Bin}(100, 0.1)$:
- $E[X_1] = 10$, $\mathrm{Var}(X_1) = 9$
- $X_2 \sim \mathrm{Poisson}(10)$:
- $E[X_2] = 10$, $\mathrm{Var}(X_2) = 10$
- Choose $X_3 \sim N(10, \sigma^2)$ with $9 < \sigma^2 < 10$, e.g. $\sigma^2 = 9.5$.
- Interpretation: When to use Binomial, Poisson, or Normal depending on context (fixed trials vs. rate processes vs. continuous/CLT approximations).

Problem 12 – Sample Mean & CLT

- For i.i.d. $X_1, \ldots, X_n$ with mean μ, variance σ^2:
- Sample mean: $\overline{X}_n = \frac{1}{n} \sum X_i$
- $E[\overline{X}_n] = \mu$
- $\mathrm{Var}(\overline{X}_n) = \sigma^2/n$
- As n increases, variance decreases (average is less variable).
- CLT (informal): standardized sum $\frac{\sum X_i - n\mu}{\sigma\sqrt{n}}$ is approximately $N(0, 1)$ for large n; hence $\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$.